

Gamma-Conditioned Bands on a Single-Name Equity (SPCX)

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July 2026

1 Setup and Benchmarks

This is a companion to *Backtest Results and Possible Improvements*, which studied static and dynamic bands on ES. Here the same band machinery is applied to **SPCX**, a Nasdaq-listed single-name equity, using the SPCX data pipeline and cost model built on the `Timmy/strategy-comparison` branch (notebook `05_spcx_extension`); the band shapes and the analysis below are from notebook `06_spcx_dynamic_bands`. The sample is 22 trading days (June 16 – July 17, 2026), 12,092 simulated SPCX option executions (JPM modeled at 1% of option flow), 1,716 five-minute bars (78 per day), with book delta *and* gamma repriced by Black–Scholes at every bar.

Two things differ materially from the ES study and drive every number below.

Units. One option contract delivers 100 shares and the hedge instrument is the stock itself (\$1 of P&L per share per \$1 move), so book delta is measured in *shares* and dollar P&L needs no contract multiplier. Delta risk is the RMS of realized residual-delta dollar P&L per bar (residual shares $\times$ \$ price move).

Cost model. The ES impact level was anchored at “ ≈ 0.5 bp per 1-lot at median ADV,” a figure that silently embeds ES’s $\approx 15\%$ volatility and its negligible clip-to-ADV ratio. Neither holds here, so the SPCX cost model is rebuilt rather than ported:

- **Square-root law, volatility-anchored.** The impact fraction of notional is $\alpha\sqrt{q/L}$ with $\alpha = Y\sigma_{\text{daily}}$ (Toth–Bouchaud prefactor $Y = 0.5$), giving $\alpha = 0.0273$ from SPCX’s median ATM implied volatility.
- **Execution-window liquidity.** Impact is charged against the liquidity available over the fill’s window, $L = \text{ADV} \times 5/390$ for an intraday

child, not against full-day ADV. ADV is real daily traded volume from Nasdaq daily bars ($\approx 60.9\text{M}$ shares).

- **Closing cross.** A fill on the final bar leans on full-day ADV, crosses at the auction price (zero spread), and takes a haircut on impact.

Over the sample SPCX traded \$123.12–\$222.80 (median \$155.27), with ATM implied volatility of 70–126% and a 20-day ADV near 60,900,000 shares, about \$9.5B a day. The measured median quoted half-spread is \$0.027 per share. The book is small relative to that liquidity: the selected strategy turns over roughly 114,000 shares a day, under 0.2% of ADV. Impact dominates spread — the winning band’s all-in cost is $\approx \$0.69/\text{share}$ against a \$0.027 half-spread — so footprint, not the quoted spread, is the binding cost. Configurations are scored by the standardized loss

$$L_{\text{std}} = \tilde{C}_{\text{exec}} + \lambda \tilde{C}_{\text{risk}}, \quad \lambda = 1,$$

normalizing cost by Full Hedge and risk by Zero Hedge.

Benchmark	Exec. cost	Risk (RMS/bar)	L_{std}
Zero Hedge (flatten at close only)	\$154,125	\$226,538	1.015
Full Hedge (re-flatten every bar)	\$10,528,317	\$0	1.000

Table 1: Benchmark bounds over the 22-day SPCX backtest. Full hedging this book against one bar’s worth of liquidity costs \$10.5M; never hedging carries \$227k of residual-delta P&L risk per bar.

2 The Gamma-Conditioned Band

Only the gamma mechanism is carried over from the ES study. The time-decay factor $\sqrt{R(t)/R(0)}$ is *dropped*: its derivation assumed the forced end-of-day flatten is expensive, and SPCX’s Nasdaq closing cross makes that flatten cheap, removing the penalty the shape was built to avoid. The band is therefore

$$|\Delta(t)| > H(t)(1 + \epsilon), \tag{1}$$

$$H(t) = \frac{H_0}{1 + c|\Gamma(t)|}, \tag{2}$$

where $\Gamma(t)$ is the book gamma repriced each bar (shares of delta regenerated per \$1 move) and $c \rightarrow 0$ recovers the static band. Note c is *not*

dimensionless in this form: it carries units of \$ per share, and the band halves at $|\Gamma| = 1/c$. This book’s median $|\Gamma|$ is 488 shares/\$, so the useful range of c is order 10^{-4} – 10^{-3} ; the grid is $c \in \{2, 5, 10, 20\} \times 10^{-4}$. (Dividing $|\Gamma|$ by its median and rescaling c gives the same family of bands with a dimensionless c .) The rationale is that delta diffuses through the band at speed $d\Delta \approx \Gamma dS$ (proposal Eq. 32): a high-gamma book breaches a fixed band faster, so tightening when $|\Gamma|$ is high keeps breach dynamics comparable across gamma regimes.

Is there signal? The band can only help if gamma varies intraday and carries information not already in $|\Delta|$. Both hold, but weakly: across bars $\rho(|\Delta|, |\Gamma|) = 0.53$ (so a good part of the gamma signal is already in delta), the within-day coefficient of variation of $|\Gamma|$ is 0.46, and the median day’s intraday max/median gamma ratio is only $1.6\times$. That $1.6\times$ is the whole range over which the band can flex, and it caps how large the effect can be.

3 Static vs. Gamma: Grid Search

Both families sweep H_0 , execution horizon and method at fixed $\epsilon = 5\%$ and $T_{\text{cooldown}} = 15$ min; the gamma family adds the tilt $c \in \{2, 5, 10, 20\} \times 10^{-4}$. The static grid runs from $H_0 = 5,000$ to 300,000 shares so both tight and wide bands are represented.

Strategy	Exec. cost	Risk/bar	L_{std}	Hedges/day
Best static	\$1.74M	\$78,303	0.5110	0.36
Best gamma	\$1.74M	\$78,303	0.5110	0.36

Table 2: Grid winners: static $H_0 = 200,000$ shares, VWAP 30min; gamma $H_0 = 300,000$ shares, $c = 2 \times 10^{-4}$, VWAP 30min. The band cuts the standardized objective by $\approx 50\%$ vs. Zero Hedge and $\approx 49\%$ vs. Full Hedge – the first-order result. Best static and best gamma are *identical* to four decimals: the winning gamma tilt is so weak ($c = 2 \times 10^{-4}$) that the band is static in all but name.

The optimum is a wide, interior band. Square-root-law impact is superlinear in clip size ($\propto q^{1.5}$), so splitting a parent hedge into n children costs $Q^{1.5}n^{-1/2}$ and the proxy charges no fixed per-ticket cost — which means one must check the optimum is not merely sitting on the tight edge of the grid. It is not. The static ladder is *U-shaped* with a clear interior minimum:

L_{std} : $H_0=5\text{k}$: 0.552, 25k: 0.569, 100k: 0.565, **200k: 0.511**, 300k: 0.712.

The best band is *wide* and fires only ≈ 0.4 times a day; tighter bands are strictly worse because impact is not negligible. The grid optimum is therefore well-behaved, and no external frequency cap or per-ticket cost is needed to make it interior.

4 Width-Matched Comparison

Setting a static band to each gamma configuration’s mean effective width (same horizon, method, ϵ , cooldown) isolates the value of *conditioning on gamma* from the value of simply being *tighter*.

H_0 (sh)	Mean width (sh)	Gamma L_{std}	Matched static L_{std}	Edge
25,000	16,520	0.5685	0.5793	+1.86%
50,000	33,039	0.5706	0.5730	+0.42%
100,000	66,079	0.5669	0.5806	+2.36%
200,000	132,158	0.5657	0.5755	+1.71%
300,000	198,236	0.5704	0.5441	−4.84%

Table 3: Width-matched gamma vs. static at $c = 10^{-3}$, TWAP 60min. Positive = gamma better.

At a matched width the gamma band is mildly risk-efficient over the widths that bind ($\approx 16\text{k}$ – 132k shares), beating an equal-width static band by +0.4% to +2.4% of standardized loss, because it holds tighter delta control exactly when convexity is high. It collapses to −4.8% at the widest setting, where the band is so wide it rarely binds and the gamma tilt merely over-tightens on the few occasions it does. So the *mechanism* is real — gamma is genuinely risk-efficient at a matched width — but it is a second-order refinement, consistent with the $1.6\times$ ceiling on intraday gamma variation, and it does *not* translate into a win at the actual optimum, which the next

section makes explicit.

5 The Operational Regime: A Hedge-Frequency Cap

A natural robustness check is whether a capacity constraint — a cap on the average number of hedges per day, the operationally binding limit for a desk pushing size through the market — changes the ranking. It does not: the optimum already fires ≈ 0.4 hedges/day, so any plausible cap is slack.

Cap/day	Static L	Gamma L	Edge	Winner
5	0.5110	0.5110	0.00%	static
10	0.5110	0.5110	0.00%	static
15	0.5110	0.5110	0.00%	static
20	0.5110	0.5110	0.00%	static
30	0.5110	0.5110	0.00%	static
none	0.5110	0.5110	+0.00%	tie

Table 4: Best static vs. best gamma under a cap on hedge frequency. Every row is identical because the wide-band optimum fires far below even the tightest (5/day) cap: the constraint never binds. Positive = gamma better.

Because the optimum hedges rarely, there is no scarce hedge budget whose allocation gamma could improve, and static wins or ties at every cap. The same holds across risk-aversion: for $\lambda \in [0.25, 4]$ the gamma band only ties or narrowly leads the static band (edge 0 to +0.37%, at $\lambda = 2$) — a marginal, not a decision-relevant, advantage.

6 Band-Edge Targeting

Replacing the full flatten with the Whalley–Wilmott-style edge target $Q_t = -\text{sign}(\Delta(t))(|\Delta(t)| - H(t))$ parks the residual at the band edge rather than round-tripping through zero.

Edge targeting is mildly positive for moderate band widths of either shape (+0.5% to +3.5%) and backfires only for the widest bands, where full-flattening into the cheap closing cross is preferable to carrying a large residual. On SPCX the effect is smaller and less systematic than on ES, where it was the strongest single refinement: the reason is the closing cross,

H_0 (sh)	10,000	25,000	50,000	100,000	200,000
Static band	+1.04%	+3.50%	+2.04%	+0.52%	−15.02%
Gamma band	+0.86%	+0.80%	+2.33%	+2.12%	−1.17%

Table 5: Improvement from edge targeting (TWAP 60min). Positive = better than full flatten.

which already delivers to every strategy the cheap carry-to-close that edge targeting is designed to buy. It is a minor tuning knob here, not a driver.

7 Cost-Model Discussion

Both cost components are size-proportional – spread (linear in shares) and square-root-law impact (superlinear, $\propto |q|^{1.5}$ once the $q\sqrt{q/L}$ form is expanded) – so both vanish as orders shrink, and the proxy has no mechanism to penalize frequent triggering on its own. That raises the question of what keeps the optimal band from collapsing to an ever-tighter, ever-more-split boundary; here the answer is that impact is large enough in absolute terms that the grid optimum is a wide interior band (Section 3), so the question is moot at the optimum. For completeness, a per-ticket sweep confirms a charge of \$10–\$100 per order is under 0.2% of execution cost and moves nothing.

Impact dominates spread throughout ($\approx$ \$0.69 vs. \$0.027 per share all-in on the winning band): this is an equity book whose binding cost is market footprint, not the quoted spread. Two caveats are specific to this sample. The 22-day window is short – one fifth of the ES sample – so the width-matched percentages should be read as indicative. And the tape draws sides i.i.d. 50/50, so option arrivals carry no predictable structure; real client flow is clustered, which would change the balance between waiting for offsetting flow and hedging.

8 Conclusion and Possible Improvements

Bottom line. On SPCX the no-trade band is the first-order win — it cuts the standardized cost–risk objective by $\approx 50\%$ versus both naive benchmarks — but **gamma conditioning is not worth its added complexity on this book**. It neither improves the grid optimum (best gamma ties best static to four decimals) nor wins under a realistic hedge-frequency cap (which

never binds). Its only residual value is a second-order, width-matched risk efficiency (+0.4 to +2.4%) that does not survive to the optimum.

Recommendation. Calibrate band *width* first — that is where essentially all the value is. Run the *static* band as the operating shape on SPCX; adopt gamma conditioning only on books where the intraday gamma range is materially larger than the $1.6\times$ seen here, or where hedging is genuinely capacity-constrained.

1. **Extend the sample.** 22 days is thin for distinguishing effects of 1–3%. Extending the SPCX window, or bootstrapping across days, would put error bars on the width-matched edge and confirm the interior optimum is stable.
2. **Volatility-conditioned band.** Gamma measures how fast delta moves per dollar of underlying; a realized-volatility divisor $1 + c_\sigma \hat{\sigma} / \tilde{\sigma}$ captures how many dollars the stock actually moves. Same plumbing, and on a name whose implied volatility spans 70–126%, the intraday variation is likely larger than the factor of 1.6 gamma offers — a more promising conditioning signal than gamma on this book.
3. **Calibrate Y and the closing-cross haircut.** The square-root prefactor ($Y = 0.5$) and the auction haircut are assumptions, and the cost levels here are large enough that they deserve TCA support.
4. **Historical VWAP weights.** VWAP currently uses same-day realized volumes (mild look-ahead); a trailing time-of-day volume profile would make it implementable.
5. **Cross-name and larger-book validation.** The gamma edge appears where hedging is expensive relative to the book. Testing a more convex, event-driven name — or JPM at a larger share of flow — would map where the gamma refinement starts to pay, which the mechanism predicts and this book does not reach.